

# MAHIKA MAHAPATRA

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• Portfolio: mahikamahapatra101.github.io/Digital-Portfolio-v2

## EDUCATION

### University of Wisconsin–Madison

Sep 2025 – May 2028 (Expected)

B.S. Computer Science | GPA: 3.40 / 4.00 | Dean's List, Spring 2026

**Relevant coursework:** Programming II & III (CS 300/400), Linear Algebra (Math 340), Digital Systems (ECE 252), Statistics (STAT 240), User Interfaces (CS 571), Discrete Mathematics (CS 240), Machine Organization and Programming (CS 354)

## TECHNICAL SKILLS

**Languages:** Java, JavaScript, HTML/CSS, Bash, R, Python, SQL, LC3, C

**Tools & frameworks:** Git/GitHub, React, FastAPI, SQLAlchemy, pandas, JUnit 5, VS Code, Eclipse, n8n, Supabase, PostgreSQL, Google Gemini / AI Agents

## PROJECTS

### RunLoad — Training Load & Injury-Risk Tracker

Jun – Jul 2026

Python · FastAPI · React · PostgreSQL · pandas | Full-Stack Web App | [github.com/mahikamahapatra101/runload](https://github.com/mahikamahapatra101/runload)

- Built a full-stack web app (Python/FastAPI, React, SQLAlchemy) that calculates Acute:Chronic Workload Ratio (ACWR), a sports-science metric, to flag when a runner's training load is rising too fast.
- Wrote a Python pipeline to clean and process 1,300+ real runs from a personal Strava export, adding a perceived-effort field the raw export didn't include.
- Hit a data loss bug after deploying to Render (SQLite doesn't persist on their free tier); migrated to Postgres on Neon to fix it, and deployed the full app live on Render and Vercel.

### ForecastGrid AI — Smart Grid Demand Forecasting

Jun 2026

R · LSTM Neural Network · Data Engineering · Time-Series Forecasting | FUSE Program, MKE Tech Hub Coalition | "In It To Win It" Award Winner

- Won the "In It To Win It" Award out of 6 competing teams at FUSE Demo Day hosted by the MKE Tech Hub Coalition.
- Cleaned and merged three real-world datasets (commercial energy, residential consumption, live Milwaukee weather feeds) in R, engineered features, and trained an LSTM neural network to predict energy demand spikes one hour ahead.
- Flagged 165 genuine high-risk demand hours out of 1,643 hours of real Milwaukee-area grid data; built a dashboard prototype delivering automated recommendations to control room operators before spikes occur.

### Stride — Campus Safety Navigation App

Mar 2026

Java · React · Dijkstra's Algorithm · Safety-Weighted Routing | CheeseHacks 24-Hour Hackathon | [github.com/mahikamahapatra101/stride-cheesehacks](https://github.com/mahikamahapatra101/stride-cheesehacks)

- Back-end engineer on a 4-person team: designed the graph data model, implemented safety-weighted Dijkstra's to prioritize well-lit paths, and delivered a clean API to the React front-end within 24 hours.
- Pivoted routing logic mid-build after initial edge weighting produced suboptimal paths; recalibrated multipliers on the fly and re-demoed a working product within the same 24 hours.

### Self-Balancing Search Tree

Jan – Mar 2026

Java · Red-Black Tree · Generics · JUnit 5 | [github.com/mahikamahapatra101/SelfBalancingSearchTree\\_RedBlackTree\\_JUnit](https://github.com/mahikamahapatra101/SelfBalancingSearchTree_RedBlackTree_JUnit)

- Implemented a generic Red-Black Tree with  $O(\log n)$  guaranteed search and insertion, handling all rotation and color-adjustment cases from scratch with no external libraries.
- Verified all RBT invariants via JUnit 5 across arbitrary object types using Java generics; used as a game leaderboard backend supporting level, playtime, and record-count sorting.

## EXPERIENCE & LEADERSHIP

### CBRE & Project Destined — AI Engineering Fellow

June – Aug 2026

n8n · Google Gemini (AI Agent) · Lovable · Supabase · PostgreSQL | [github.com/mahikamahapatra101/MultiGuard-AI](https://github.com/mahikamahapatra101/MultiGuard-AI)

- Selected for an applied AI fellowship in partnership with CBRE, focusing on LLM workflow automation and commercial real estate underwriting.
- Built an n8n and Google Gemini workflow to parse complex multifamily Offering Memorandums, extracting core financial metrics (NOI, cap rate, occupancy, debt yield) with strict schema constraints.
- Coded a deterministic risk engine enforcing institutional investment thresholds (90% occupancy floor, 8% debt yield minimum) to automatically flag covenant breaches and output Go/No-Go verdicts.
- Connected a React frontend (built via Lovable) to a Supabase PostgreSQL database via secure webhooks to log deal audit trails and export PDF investment memos.

### Ground Truth Cafe — UW–Madison | Barista

Jan 2026 – Present

- Maintained reliable performance and accuracy in a high-volume cafe while carrying a full CS courseload.

### Science National Honor Society | Co-Founder, Sussex Hamilton High School

2024

- Founded the school's first-ever SNHS chapter: designed the application process, led a 6-month recruitment effort growing to 40+ members, and built all chapter programming with no existing template.

## AWARDS & ACTIVITIES

- FUSE Program: "In It To Win It" Award Winner, MKE Tech Hub Coalition (2026)
- NCWIT Aspirations in Computing: State Winner 2025; State Honorable Mention 2022, 2023, 2024